Smart Door Lock System Project Report

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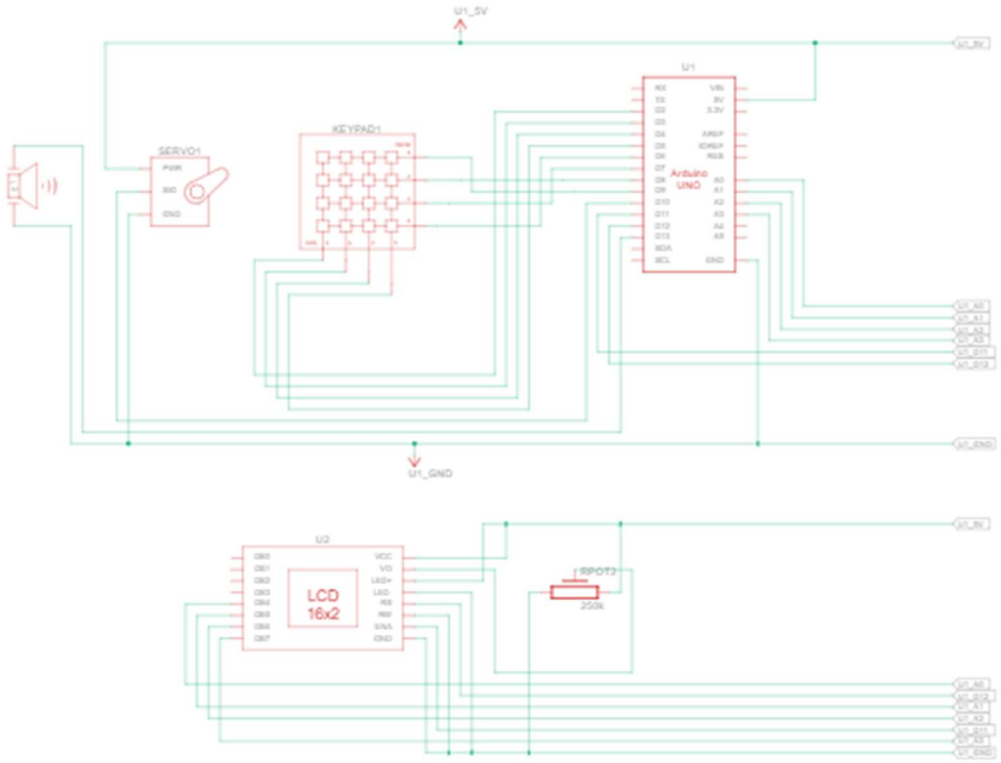
Abstract:

This project presents the design and implementation of a password-based smart door lock system using an embedded platform. The system enhances security by replacing traditional locks with a digital authentication mechanism. A user enters a predefined password through an input device, and the system verifies it using a microcontroller. If the password is correct, the door unlocks; otherwise, access is denied. The system also provides feedback through display or indicators. This project demonstrates the practical application of embedded systems in home and office security.

Components Required:

- Arduino UNO
- 4×4 Keypad
- LCD 16×2
- Servo motor
- Buzzer
- Potentiometer (10k)
- Breadboard

Circuit Diagram:



Working Principle:

1. **Power Supply ON**
   The system is powered using a DC supply, and all components (microcontroller, keypad, display) are initialized.

2. **System Initialization**
   The microcontroller loads the predefined password stored in memory and sets the input/output ports.

3. **User Input**
   The user enters a password using the keypad.

4. **Data Processing**
   The entered password is read and stored temporarily by the microcontroller.
5. **Password Verification**
   The system compares the entered password with the stored password.
6. **Access Decision**
   - If the password is **correct** → proceed to unlock
   - If the password is **incorrect** → deny access
7. **Door Unlocking**
   When the password is correct, the microcontroller sends a signal to the motor/servo, which opens the door.
8. **Display Feedback**
   The system displays messages like:
   - "Access Granted"
   - "Access Denied"
9. **Security Handling**
   After multiple wrong attempts, the system can trigger a buzzer or temporarily block input.
10. **Auto Reset / Relock**
    After a few seconds, the door automatically locks again and the system resets for the next user.

Code:

```
#include <Keypad.h>

#include <LiquidCrystal.h>

#include <Servo.h>


// LCD Pins (RS, EN, D4, D5, D6, D7)

LiquidCrystal lcd(12, 11, A0, A1, A2, A3);


// Servo

Servo lockServo;


// Buzzer Pin

int buzzer = 13;


// Password

String password = "1234";

String input = "";


// Keypad setup

const byte ROWS = 4;
```

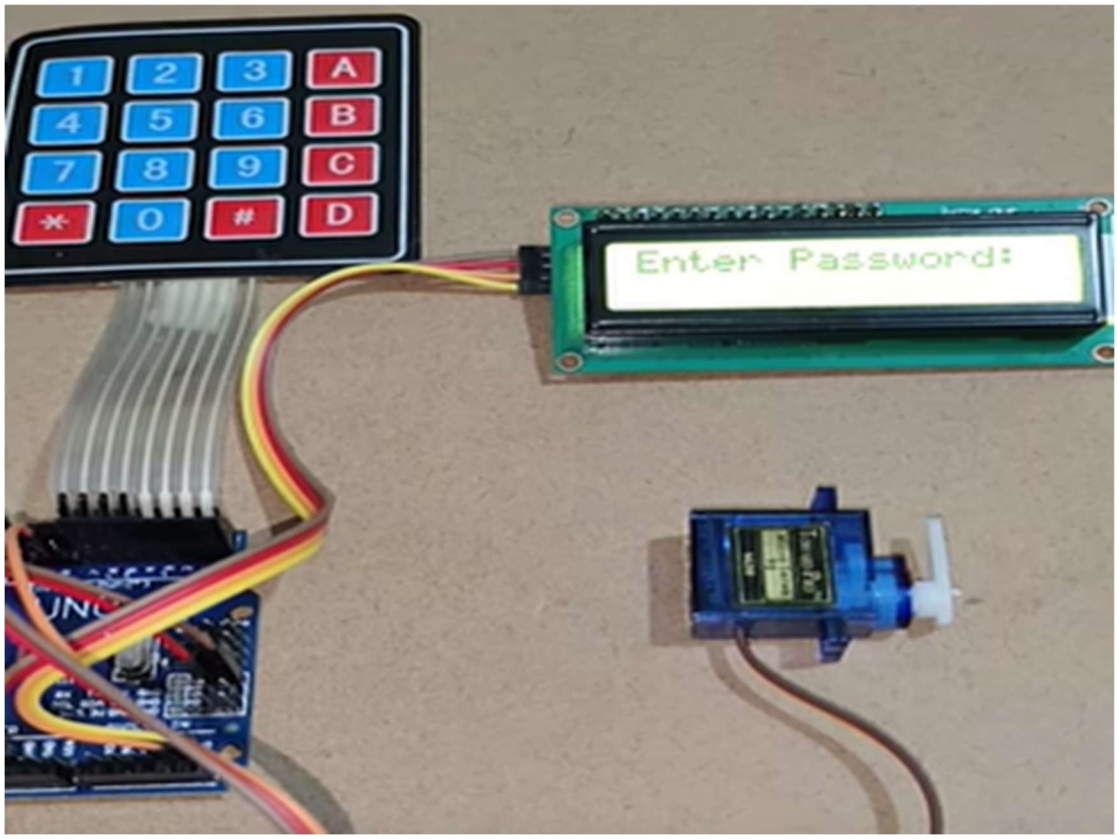
```
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
COLS);

void setup() {
  lcd.begin(16, 2);
  lockServo.attach(10);
  pinMode(buzzer, OUTPUT);

  lcd.print("Smart Lock");
```

```
  delay(2000);
  lcd.clear();

  lockServo.write(0); // Locked position
}

void loop() {
  lcd.setCursor(0, 0);
  lcd.print("Enter Password:");

  char key = keypad.getKey();

  if (key) {
    if (key == '#') {
      // Check password
      lcd.clear();
      if (input == password) {
        lcd.print("Access Granted");
        lockServo.write(90); // Unlock
        delay(5000);
        lockServo.write(0);  // Lock again
```

```
  } else {
    lcd.print("Wrong Password");
    digitalWrite(buzzer, HIGH);
    delay(1000);
    digitalWrite(buzzer, LOW);
  }
  input = "";
  delay(2000);
  lcd.clear();
}
else if (key == '*') {
  input = "";
  lcd.clear();
  lcd.print("Cleared");
  delay(1000);
  lcd.clear();
}
else {
  input += key;
  lcd.setCursor(0, 1);
  lcd.print(input);
```

```
    }
  }
}
```

Output:



Applications:

1 **Home Security Systems**

Used in houses to provide secure and keyless entry, reducing the risk of unauthorized access.

2 **Office Security**
Helps protect confidential areas by allowing access only to authorized personnel.

3 **Smart Buildings**
Integrated into modern smart homes and buildings for automated and remote-controlled access.

4 **Hotels and Hostels**
Used for room access control instead of traditional keys.

5 **Bank Lockers and Secure Rooms**
Provides additional security layers for sensitive storage areas.

Conclusion:

This project also provides a strong understanding of microcontroller programming, hardware interfacing, and real-time system design. Furthermore, it can be extended with advanced features such as biometric authentication, mobile app control, and IoT integration for improved functionality and remote access.